

The Smagorinsky and Kolmogorov Constants

A unified spectral theory, two-point closures and a three-language numerical experiment: the closed-form relation $C_s = 1/(\pi(3C_K/2)^{3/4})$, its filter-family corrections, Monte-Carlo verification on synthetic K41 fields and a priori tests on DNS turbulence.

$C_s(C_K = 1.50) = 0.17327$ — Lilly 1966: 0.17326

wild8highlander

research_col_smar · programs P1-P4 · Python · C++ · Julia
monograph edition: English